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READING THE DATASHEETS OF ELECTRONIC DEVICES

Designers have a large variety of operational amplifiers (op-amps) to choose from. So we focus this month on op-amps to help designers decide which category is best suited for a given set of applications

■ ANIL KUMAR MAINI AND
VARSHA AGRAWAL

Depending upon the performance specifications and consequently their application areas, and also on the basis of their internal circuit, op-amps are broadly classified as general-purpose op-amps, high-speed and high-bandwidth op-amps, op-amp comparators, Norton op-amps, instrumentation op-amps and isolation op-amps.

Like other semiconductor devices, op-amps too are characterised by a wide range of performance specifications. Again, the importance and relevance of a given performance parameter to a certain application circuit could be more or less than the other parameters.

Selection criteria

Choosing the right op-amp for a given application is usually a three-step process. First, appreciate relevant performance parameters and the level of these parameters required for the intended application. Second, identify the category of op-amps that the desired device is likely to belong to. Next, select the type number from the device catalogues of the relevant manufacturers.

For example, if you are looking for an op-amp to be used in a trans-impedance amplifier configuration to convert the photo-current resulting from a laser radiation pulse falling on a photodiode into an equivalent voltage signal, first

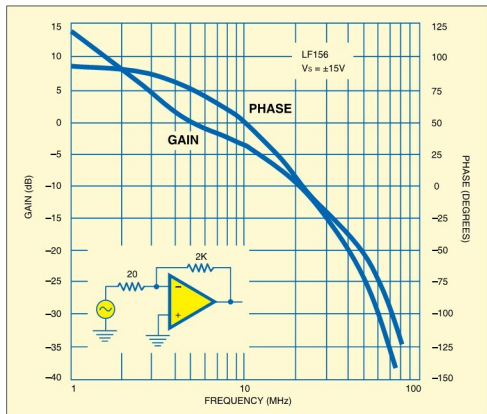


Fig. 1: Frequency response of LF156

identify relevant parameters such as gain bandwidth, slew rate and trans-impedance gain available at the desired bandwidth and then fix the desired values for these parameters. Having done that, see if any of the available general-purpose op-amps meets the required performance specifications or you need a high-speed, high-bandwidth op-amp.

Op-amps are characterised by performance specifications like supply voltage range, power dissipation, operating temperature range, open-loop gain, gain bandwidth product,

common-mode rejection ratio (CMRR), power supply rejection ratio (PSRR), input impedance, slew rate, offsets, offset drifts and noise-related specifications.

Open-loop gain is the ratio of single-ended output to differential input. It influences gain accuracy when the op-amp is wired as an amplifier. Higher open-loop gain is desirable as it gives higher gain accuracy. It is, in fact, the loop gain—ratio of open-loop gain to the closed-loop gain—which decides the gain accuracy.

Bandwidth represents the fre-